

VertiCore Enterprise Agent Platform

Complete Architecture Specification

1. Executive Summary

VertiCore is an enterprise-grade AI agent orchestration platform designed for regulated industries such as finance and healthcare. It enables the safe deployment of intelligent, multi-step, tool-using AI agents while maintaining strict guarantees around determinism, auditability, privacy, and compliance.

Unlike chatbot-style systems or autonomous agent swarms, VertiCore separates **reasoning**, **execution**, and **governance** into distinct layers. This allows powerful AI capabilities without sacrificing control or trust.

2. Core Design Principles

1. **Durable Execution Over Ephemeral Reasoning**
 2. **Separation of Concerns (Control vs Intelligence)**
 3. **Policy-Governed Autonomy**
 4. **PII-Safe by Construction**
 5. **Deterministic, Replayable Workflows**
 6. **Framework Optionality (LangGraph / LangChain)**
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3. High-Level System Architecture

```
Client / UI
  ↓ GraphQL
API Gateway
  ↓ gRPC
Control Plane (Go)
  ↓ Start Workflow
Temporal Orchestrator
  ↓ Activity Tasks
Python Intelligence Plane
  ├── PII Services
  ├── Agent Reasoning (LangGraph)
  ├── Tool Execution
  └── LLM Gateway
  ↓
Audit & Persistence Layer
```

4. Control Plane (Go)

Responsibilities

- Session management
- Workflow initiation
- Authorization & policy checks
- Routing and orchestration

Key Components

- GraphQL Gateway (Apollo)
- Control Service
- Open Policy Agent (OPA)

The Control Plane is deterministic, low-latency, and stateless.

5. Temporal Orchestration Layer

Temporal is the execution backbone of VertiCore.

Guarantees

- Exactly-once execution
- Automatic retries
- Long-running workflows
- Deterministic replay

Temporal owns **workflow state** and agent coordination.

6. Intelligence Plane (Python)

Responsibilities

- PII sanitization (Presidio)
- Agent reasoning (LangGraph / LangChain)
- Retrieval-Augmented Generation (Qdrant)
- LLM inference (LiteLLM)

The Intelligence Plane is **not** trusted with persistence or policy enforcement.

7. Secure Agent Execution Model

PII Sandwich Protocol

```
Raw Input
  → Redaction
  → Tokenized Input
  → Agent Reasoning
  → Tokenized Output
  → Rehydration
```

LLMs never see raw PII.

8. Agent Reasoning with LangGraph

LangGraph is used as an **intra-agent reasoning engine**.

Characteristics

- Static graphs
- Deterministic routing
- Constrained LLM outputs

LangGraph never: - Calls tools directly - Persists memory - Coordinates other agents

9. Human-in-the-Loop Semantics

Human review is a **workflow concern**, not an agent concern.

Pattern

```
Agent requests review
↓
Temporal suspends workflow
↓
Human acts (minutes-days)
↓
Workflow resumes
↓
Fresh agent execution
```

Agent memory is never resumed.

10. Multi-Agent Workflow Architecture

Core Rule

Agents do not call agents. Workflows coordinate agents.

Patterns

Sequential (Pipeline)

Mortgage underwriting

Parallel (Fan-Out)

Fraud detection

Hierarchical (Supervisor)

Clinical case review

Human-in-the-Loop

Insurance claims

Temporal coordinates all patterns.

11. Memory Architecture

Memory Layers

1. Workflow Memory (Temporal)
2. Agent Working Memory (LangGraph)
3. Long-Term Knowledge (Vector DB)
4. Audit Memory (PostgreSQL)

Memory promotion is explicit and policy-governed.

12. Security & Governance

Controls

- OPA policy enforcement
- Tool allowlists

- RBAC-filtered retrieval
- Immutable audit logs

Threats Mitigated

- Prompt injection
 - Data exfiltration
 - Rogue agent behavior
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13. Audit & Observability

- OpenTelemetry tracing
- Node-level audit logs
- End-to-end trace IDs

Full forensic replay is supported.

14. Example Enterprise Agents

Mortgage Underwriting Agent

Fraud Detection Agent

Clinical Documentation Review Agent

Insurance Claim Triage Agent

Each agent runs inside a governed workflow.

15. Developer Onboarding

Steps

1. Define workflow
 2. Design static graph
 3. Constrain routing
 4. Declare tools & policies
 5. Test with replay
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16. Platform Comparison

Feature	VertiCore	AWS AgentCore	DIY LangChain
Determinism	✓	⚠	✗
Compliance	✓	⚠	✗
Long-running	✓	⚠	✗

17. Why VertiCore

VertiCore enables intelligent agents that are **powerful yet governable**, suitable for mission-critical financial and healthcare systems.

18. Key Takeaways

VertiCore treats agents as **deterministic components inside durable workflows**, not autonomous actors. This design uniquely enables safe enterprise AI at scale.